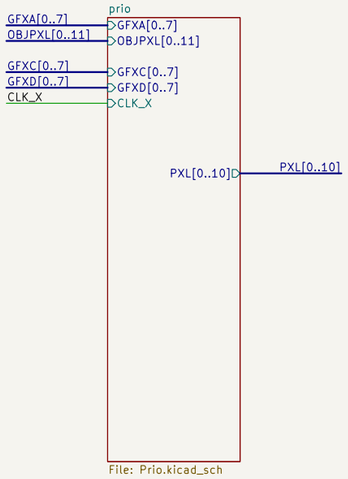
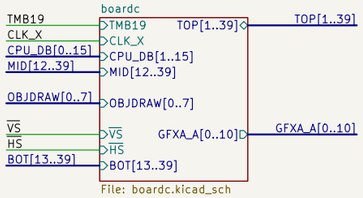
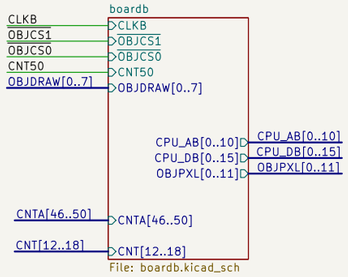
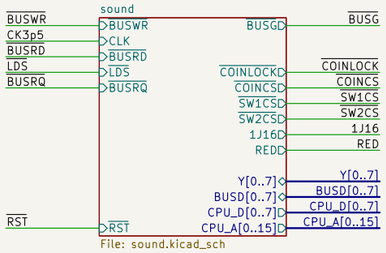
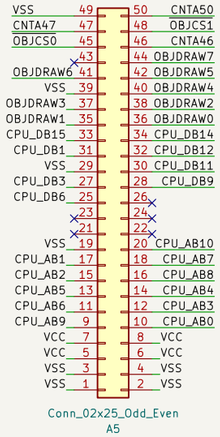


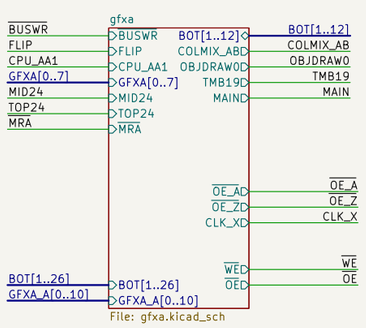


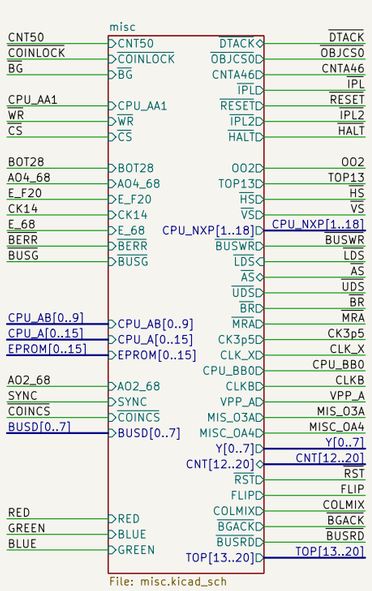


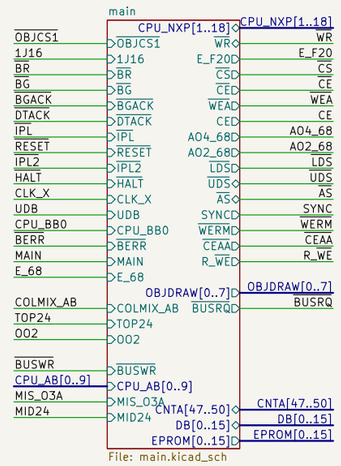












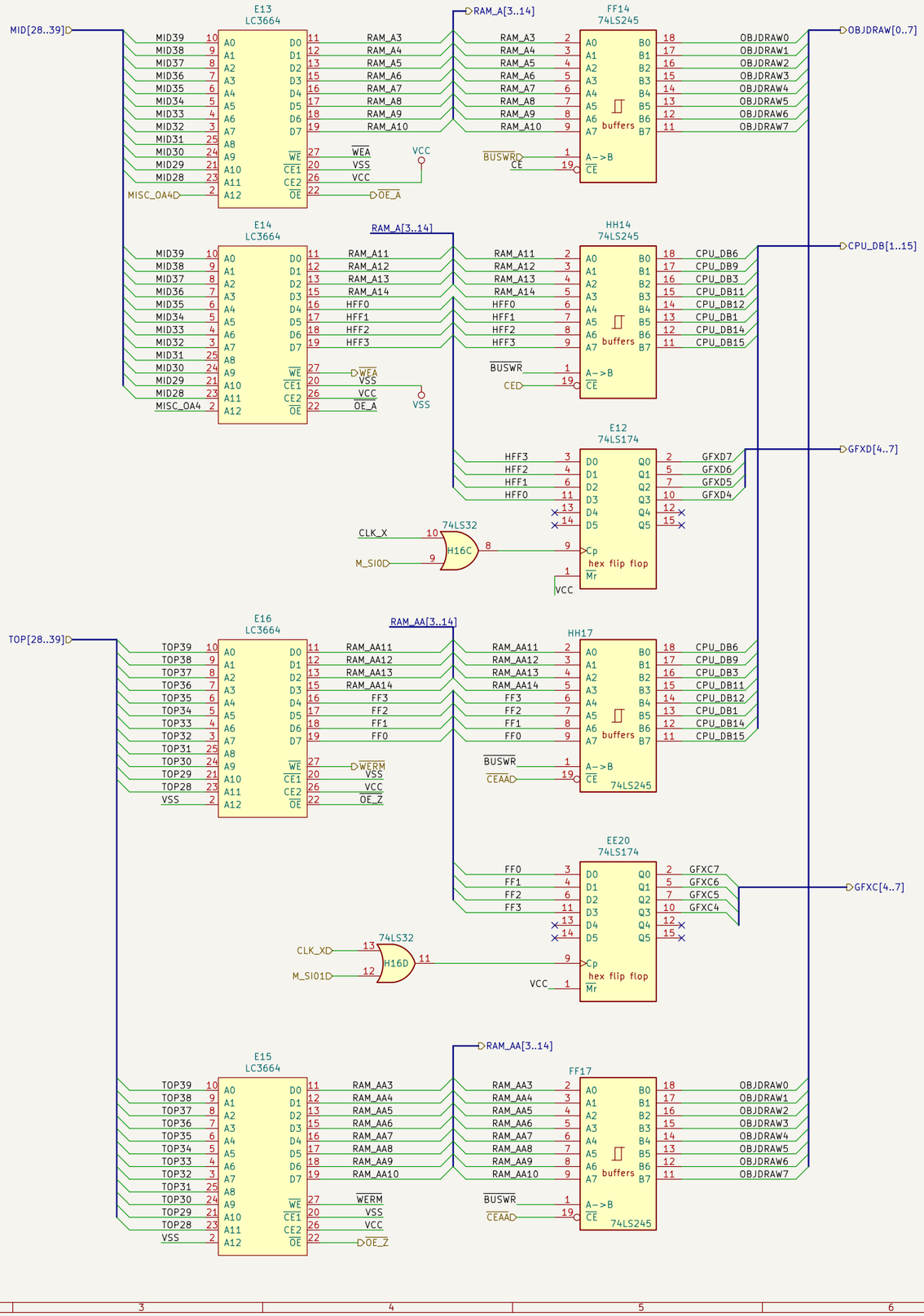


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File: ktiger.kicad_sch

Title: Kyukyoku Tiger –Bootleg–

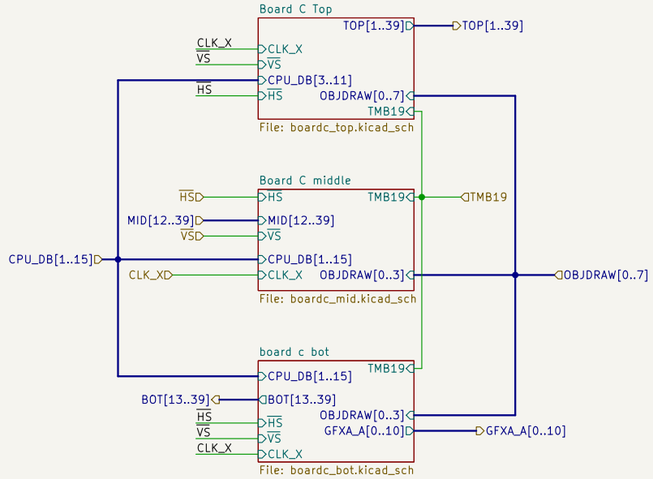
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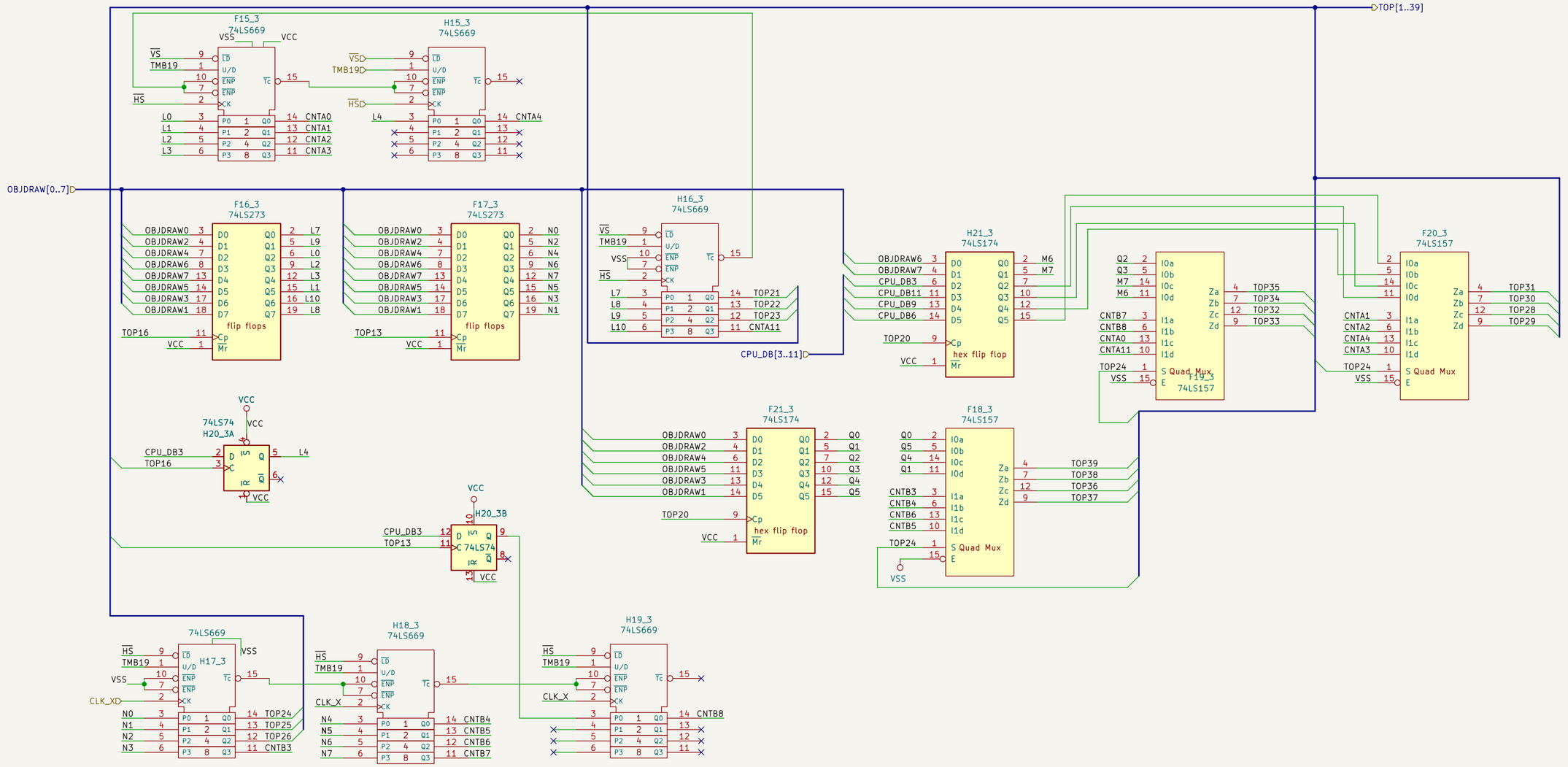


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Title: Kyukyoku Tiger –Bootleg–

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F15_3
74LS669
VSS
VCC
H15_3
74LS669
OBJDRAW[0..7]
TOP[1..39]
F16_3
74LS273
flip flops
F17_3
74LS273
flip flops
H16_3
74LS669
CPU_DB[3..11]
H21_3
74LS174
hex flip flop
F20_3
74LS157
S Quad Mux
F19_3
74LS157
S Quad Mux
F21_3
74LS174
hex flip flop
F18_3
74LS157
S Quad Mux
74LS74
H20_3A
H20_3B
74LS669
H17_3
H18_3
74LS669
H19_3
74LS669
JOTEGO
JOTEGO
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File: boardc_top.kicad_sch
Title:
Size: A3
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MID[12..39]

F10_3
74LS669

HS
TMB19
VSS
CLK_XD
L0
L1
L2
L3
MID26
MID25
MID24
CNTA3

F9_3
74LS669

HS
TMB19
CLK_X
L4
L5
L6
L7
CNTA4
CNTA5
CNTA6
CNTA7

CPU_DB[1..15]

H11_3
74LS273

CPU_DB14
CPU_DB12
CPU_DB3
CPU_DB6
CPU_DB9
CPU_DB11
CPU_DB1
CPU_DB15
MID16
VCC
N6
N4
N2
N0
N1
N3
N5
N7
flip flops

F11_3
74LS273

CPU_DB6
CPU_DB3
CPU_DB12
CPU_DB14
CPU_DB15
CPU_DB1
CPU_DB11
CPU_DB3
MID13
VCC
L0
L2
L4
L7
L5
L6
L3
L1
flip flops

F8_3
74LS669

HS
TMB19
CLK_X
CNTA8

MID[12..39]

H14_3
74LS174

CPU_DB14
CPU_DB15
OBJDRAW3
OBJDRAW2
OBJDRAW1
OBJDRAW0
MID20
VCC
M6
M7
hex flip flop

H12_3
74LS157

Q4
Q3
M7
M6
CNTA7
CNTA8
CNTB3
CNTB4
MID24
VSS
MID35
MID34
MID32
MID33
S Quad Mux

CNTB5
CNTB6
CNTB7
CNTB8
MID30
MID31
MID28
MID29

OBJDRAW[0..3]

CPU_DB[1..15]

OBJDRAW[0..3]

H13A
74LS74

VCC
OBJDRAW0
MID16

H13B
74LS74

VCC
OBJDRAW0
MID13

F14_3
74LS174

CPU_DB6
CPU_DB9
CPU_DB3
CPU_DB12
CPU_DB11
MID20
VCC
Q0
Q1
Q2
Q3
Q4
Q5
hex flip flop

F13_3
74LS157

CNTA3
CNTA4
CNTA6
CNTA5
MID24
VSS
MID39
MID38
MID36
MID37
S Quad Mux

H10_3
74LS669

VS
TMB19
VSS
HS
N0
N1
N2
N3
MID23
MID22
MID21
CNTB3

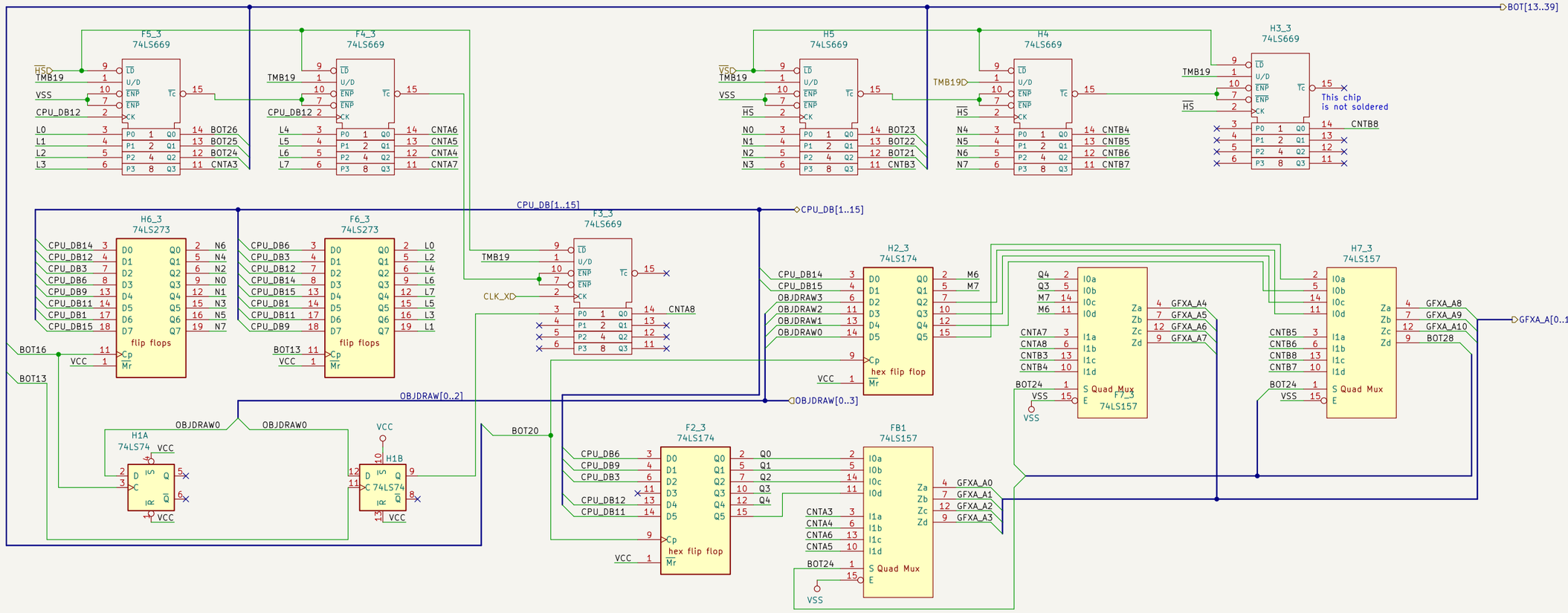
H9_3
74LS669

VS
TMB19
HS
N4
N5
N6
N7
CNTB4
CNTB5
CNTB6
CNTB7

H8_3
74LS669

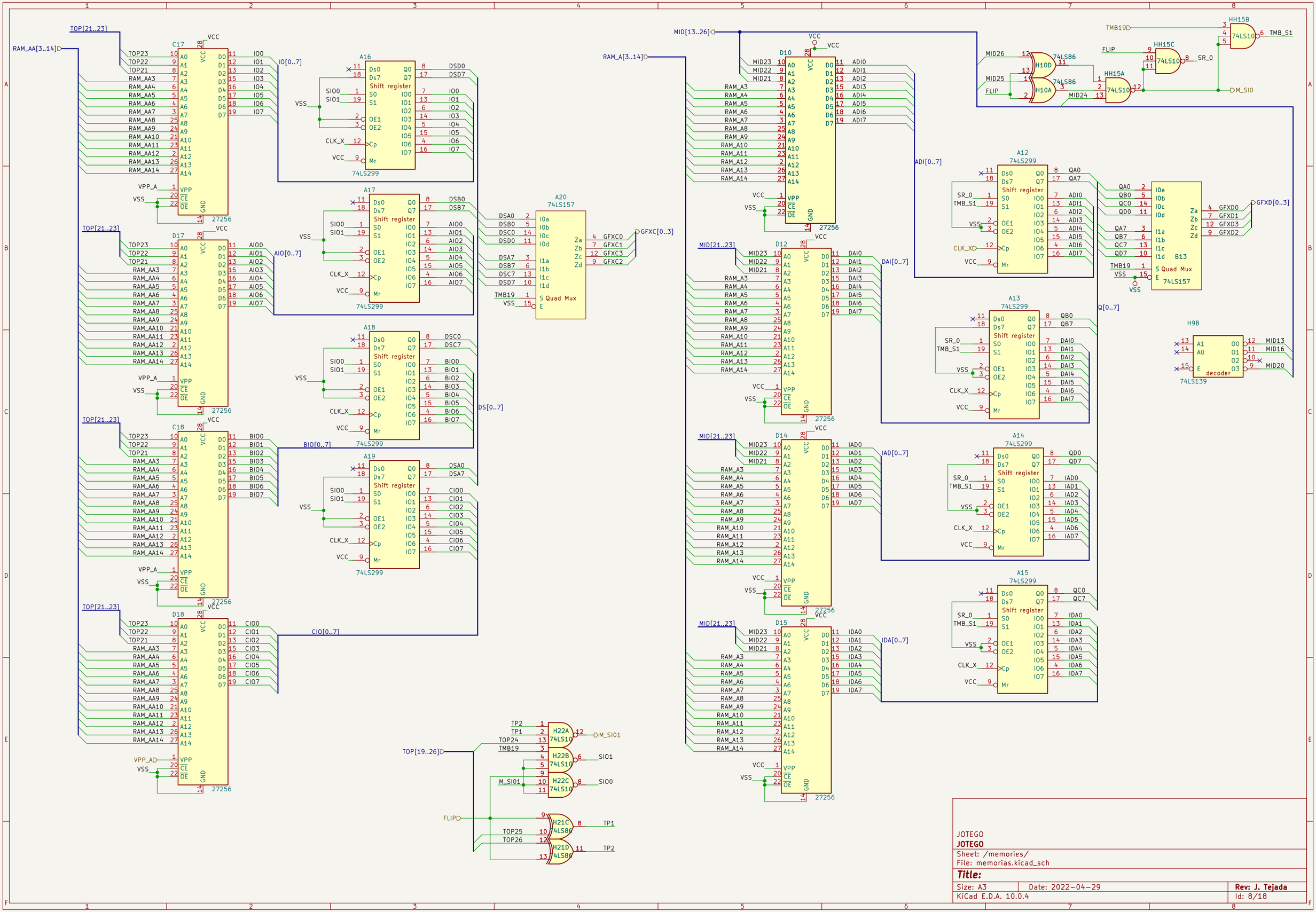
VS
TMB19
HS
CNTB8

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BOT[1..26]
GFXA_A[0..10]
H9A
74LS139
decoder
BOT13
BOT16
BOT20
EG9
CXK5816
EG10
CXK5816
GFXA_A0
GFXA_A1
GFXA_A2
GFXA_A3
GFXA_A4
GFXA_A5
GFXA_A6
GFXA_A7
GFXA_A8
GFXA_A9
GFXA_A10
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
VSS
WE
OE
D0
D1
D2
D3
D4
D5
D6
D7
CE
BOT23
BOT22
BOT21
VCC
27512
B10
GND
BOT[21..26]
A10
74LS299
Shift register
I00
I01
I02
I03
I04
I05
I06
I07
Ds0
Ds7
S0
S1
OE1
OE2
Cp
Mr
CLK_X
I1A
I0B
I1B
I0A
I1D
I0D
A11
74LS157
S Quad Mux
I0a
I0b
I0c
I0d
I1a
I1b
I1c
I1d
Za
Zb
Zc
Zd
GFXA0
GFXA1
GFXA2
GFXA[0..7]
FLIP
J15B
74LS04
TMB19
BOT26
BOT25
BOT24
FLIPD
CPU_AA1
H10B
H10C
74LS86
74LS10
F9C
F10C
74LS32
F9B
S1
TMB19
F9A
S0
COLMIX_AB
D1A
CLK_X
D[0..7]
E11
74LS174
hex flip flop
GFXA7
GFXA6
GFXA5
GFXA4
GFXA3
GFXA[0..7]
FG11
74LS175
MRA
F10A
74LS32
F10B
F10D
WE
MAIN
BUSWR
TOP24
MID24
H11B
H11C
H11D
74LS02
OE_A
OE_Z
OE
BOT24
A8
A9
A10
FR7
A103J
HR7
VCC
OBJDRAW0
BOT11
BOT12
FG7
HG7
buffers
74LS245
A->B
A[0..10]
BOT[1..26]
BOT1
BOT2
BOT3
BOT4
BOT5
BOT6
BOT7
BOT8
JOTEGO
JOTEGO
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TOP[21..23]
RAM_AA[3..14]
C17
27256
IO[0..7]
A16
74LS299
A17
A20
74LS157
GFXC[0..3]
D17
AIO[0..7]
A18
DS[0..7]
C18
BIO[0..7]
A19
D18
CIO[0..7]
MID[13..26]
RAM_A[3..14]
D10
ADI[0..7]
A12
74LS299
B13
GFXD[0..3]
Q[0..7]
A13
74LS299
H9B
74LS139
MID[21..23]
D12
DAI[0..7]
D14
IAD[0..7]
A14
74LS299
D15
IDA[0..7]
A15
74LS299
TOP[19..26]
H22A
74LS10
H22B
74LS10
H22C
74LS10
M_SIO1
SIO1
SIO0
FLIP
H21C
74LS86
H21D
74LS86
TP1
TP2
H10D
74LS86
H10A
74LS86
HH15A
74LS10
HH15B
74LS10
HH15C
74LS10
TMB19
TMB_S1
SR_0
M_SIO
JOTEGO
JOTEGO
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Size: A3
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OBJDRAW

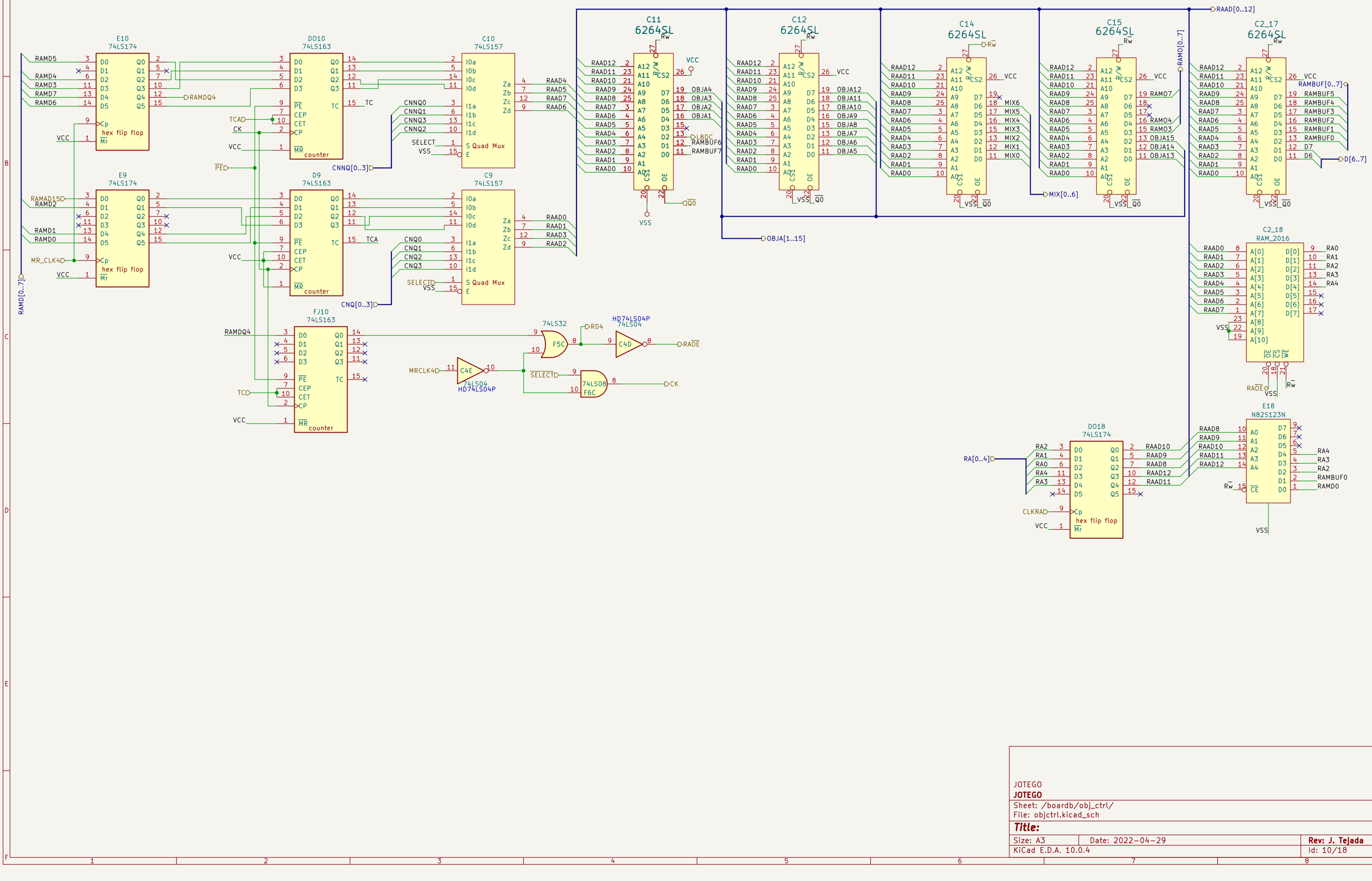
SELECT
74LS74
E5A
VSS
PE
R
F11B
74LS74
VCC
RAMD7
MR_CLK3
F11A
74LS74
VCC
D11
74LS669
SN74LS669N
VSS
CK
TC
D012
74LS367
OBJA1
OBJA2
OBJA4
OBJA3
LBDC
Q0
OEa
OEb
OBJA[0..15]
VCC
F14
27512
OBJA0
OBJA1
OBJA2
OBJA3
OBJA4
OBJA5
OBJA6
OBJA7
OBJA8
OBJA9
OBJA10
OBJA11
OBJA12
OBJA13
OBJA14
OBJA15
OBJR0
OBJR1
OBJR2
OBJR3
OBJR4
OBJR5
OBJR6
OBJR7
VSS
FJ3
27512
OBJQ0
OBJQ1
OBJQ2
OBJQ3
OBJQ4
OBJQ5
OBJQ6
OBJQ7
F17
27512
OBJU0
OBJU1
OBJU2
OBJU3
OBJU4
OBJU5
OBJU6
OBJU7
VCC
OBJA10
OBJA1
OBJA2
OBJA3
OBJA4
OBJA5
OBJA6
OBJA7
OBJA8
OBJA9
OBJA10
OBJA11
OBJA12
OBJA13
OBJA14
OBJA15
F015
27512
OBJS0
OBJS1
OBJS2
OBJS3
OBJS4
OBJS5
OBJS6
OBJS7
VSS
CPU_DB[0..15]
EO7
74LS245
CPU_DB6
CPU_DB9
CPU_DB3
CPU_DB11
CPU_DB12
CPU_DB1
CPU_DB15
CPU_DB14
RAMAD8
RAMAD9
RAMAD10
RAMAD11
RAMAD12
RAMAD13
RAMAD14
RAMAD15
buffers
A->B
bRD
OBJC5
D013
74LS374
OBJA5
OBJA6
OBJA7
OBJA8
OBJA9
OBJA10
OBJA11
OBJA12
3-state
flip flops
Q0
D014
74LS374
MIX0
MIX1
MIX2
MIX3
MIX4
MIX5
MIX6
MRCLK[0..5]
EO4
74LS138
MRCLK0
MRCLK5
MRCLK1
address
decoder
MR_CLK1
MR_CLK2
MR_CLK3
MR_CLK4
SELECT
R
CLKCNT
MR_CLK[1..4]
D015
74LS374
OBJA13
OBJA14
OBJA15
RAMO3
RAMO4
RAMD0
RAMD1
RAMD2
RAMD3
RAMD4
MR_CLK1
D016
74LS374
RAMO7
D6
D7
D[6..7]
RAMD0
RAMD2
RAMD3
MR_CLK2
RAMD[0..15]
VCC
F13
74LS32
OBJCS
OBJCS1
OBJCS0
HO19C
HO18C
CLKBUF
CNTA46
CNTA47
HD74LS04P
74LS04
C4F
74LS08
FF1D
CNTA50
VSS
OBJQ0
OBJQ1
OBJQ2
OBJQ3
OBJQ4
OBJQ5
OBJQ6
OBJQ7
CLKBUF
CLKB
VCC
HO17
74LS166
OBJR0
OBJR1
OBJR2
OBJR3
OBJR4
OBJR5
OBJR6
OBJR7
HO14
74LS166
OBJS0
OBJS1
OBJS2
OBJS3
OBJS4
OBJS5
OBJS6
OBJS7
HO16
74LS166
F7C
74LS32
Q3
CLKB
F4
74HC164
DSA
DSB
CP
74LS164
CNT50
CNT18
CNT46
CNT17
F6A
74LS08
F5B
74LS32
E6B
74LS08
HO15B
74LS74
RAMO7
UD
HO15A
74LS74
CNT18
OEA
OEB
74LS08
F20_2A
F20_2B
F20_2D
F20_2C
F18B
F18C
F18D
F18A
HO19A
74LS32
H20A
74LS08
WE
OE
F21_2
74LS244
PXLB[0..31]
PXLA[0..31]
74LS32
HO18D
HO18B
HO18A
H24A
74LS20
74F20
H3A
HD74LS04P
74LS04
F6B
F6D
F6A
F6C
CLKRA
CLKCNT
CNT12
H3F
HD74LS04P
CLKRA
QQ2
QQ0
Q0
Q1
OBJA0
Q2
Q3
EO2
74LS175
EO1
74LS175
DO1
74LS163
counter
CLKB
CLKCNT
C4A
HD74LS04P
MRCLK3
F5A
74LS32
A03
MRCLK2
C4C
C4B
HD74LS04P
SELECT
CNT50
CNT18
CNT46
CNT17
F6A
74LS08
F5B
74LS32
E6B
74LS08
74LS74
74LS74
74LS74
D5A
74LS393
82S126-9
DO4
DO3
EO3
hex flip flop
74LS174
MRCLK0
MRCLK1
MRCLK2
MRCLK3
MRCLK4
MRCLK5
MRCLK[0..5]
74LS393
74LS393
74LS393
74LS393
A00
A01
A02
A03
A04
A05
A06
A07
C6
C7
C8
Quad Mux
74LS157
RAM0
RAM1
RAM2
RAM3
RAM4
RAM5
RAM6
RAM7
RAM8
RAM9
RAM10
DO8
DO6
RAM_2016
RAMD[0..15]
OBJCSAB
CPU_AB0
CPU_AB9
CPU_AB6
CPU_AB3
CPU_AB4
CPU_AB5
CPU_AB2
CPU_AB8
bRD
CPU_AB10
CPU_AB[0..10]
SELECT
VSS
RAMD[0..7]
RAMAD15
MR_CLK4
MRCLK4
TC
TCA
SELECT
Q0
RA[0..4]
CLKRA
RW
SELECT
obj_ctrl
File: objctrl.kicad_sch
CNQ[0..3]
CNNQ[0..3]
LBDC
line_counter
dump[0..7]
CNT18
CNT17
OEB
AB8
AA8
CLKB
CLKCNT
OEA
CET
File: linecounter.kicad_sch
line_buffer
BUFA[0..7]
PXLA[0..11]
PXLB[0..11]
OBJPXL[0..11]
MIX[0..6]
WE
OE
CLKBUF
D[6..7]
File: linebuffer.kicad_sch
OBJDRAW[0..7]
OBJDRAW0
OBJDRAW1
OBJDRAW2
OBJDRAW3
OBJDRAW4
OBJDRAW5
OBJDRAW6
OBJDRAW7
buffers
A->B
bRD
OBJC5
EO8
74LS245
RAMD0
RAMD1
RAMD2
RAMD3
RAMD4
RAMD5
RAMD6
RAMD7
D017
74LS374
RAMBUF0
RAMBUF1
RAMBUF2
RAMBUF3
RAMBUF4
RAMBUF5
RAMBUF6
RAMBUF7
3-state
flip flops
MR_CLK3
D19
74LS169
BUFA6
BUFA5
BUFA4
BUFA3
U/D
CEP
CET
CLKB
BUFA[0..7]
E19
74LS169
BUFA0
BUFA1
BUFA2
BUFA7
CET
CNTA[46..50]
OBJDRAW[0..7]
CPU_DB[0..15]
CPU_AB[0..10]
CNT[12..18]
OBJPXL[0..11]
CNT50
CNTA47
CNTA46
OBJCS1
OBJCS0
OBJDRAW6
OBJDRAW3
OBJDRAW1
CPU_DB15
CPU_DB1
CPU_DB3
CPU_DB6
CPU_AB10
CPU_AB7
CPU_AB8
CPU_AB4
CPU_AB3
CPU_AB0
CPU_AB1
CPU_AB2
CPU_AB5
CPU_AB6
CPU_AB9
A21_Conn_2x25_Odd_Even
CNT18
CNT12
CLKBD
B14_Conn_2x25_Odd_Even
OBJPXL9
OBJPXL7
OBJPXL5
OBJPXL3
OBJPXL1
CNT17
CNT[12..18]
OBJPXL[0..11]
VCC
VSS

JOTEGO
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File: boardb.kicad_sch

Title: Kyukyoku Tiger –Bootleg–

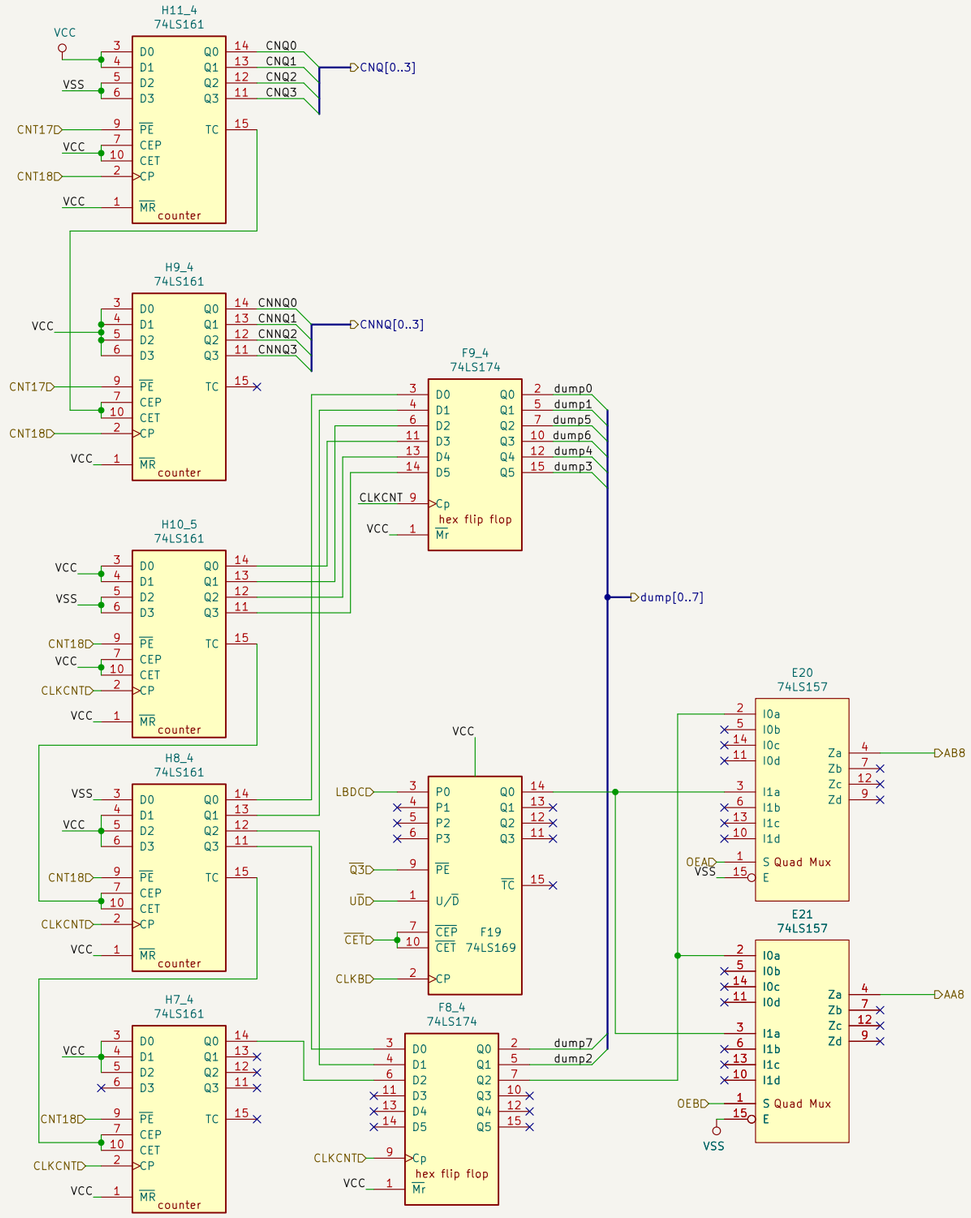
Size: A2	Date: 2022-04-29	Rev: J. Tejada
KiCad E.D.A. 10.0.4		Id: 9/18

OBJCTRL



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Sheet: /boardb/obj_ctrl/		
File: objctrl.kicad_sch		
Title:		
Size: A3	Date: 2022-04-29	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 10/18

Line buffer dump counter

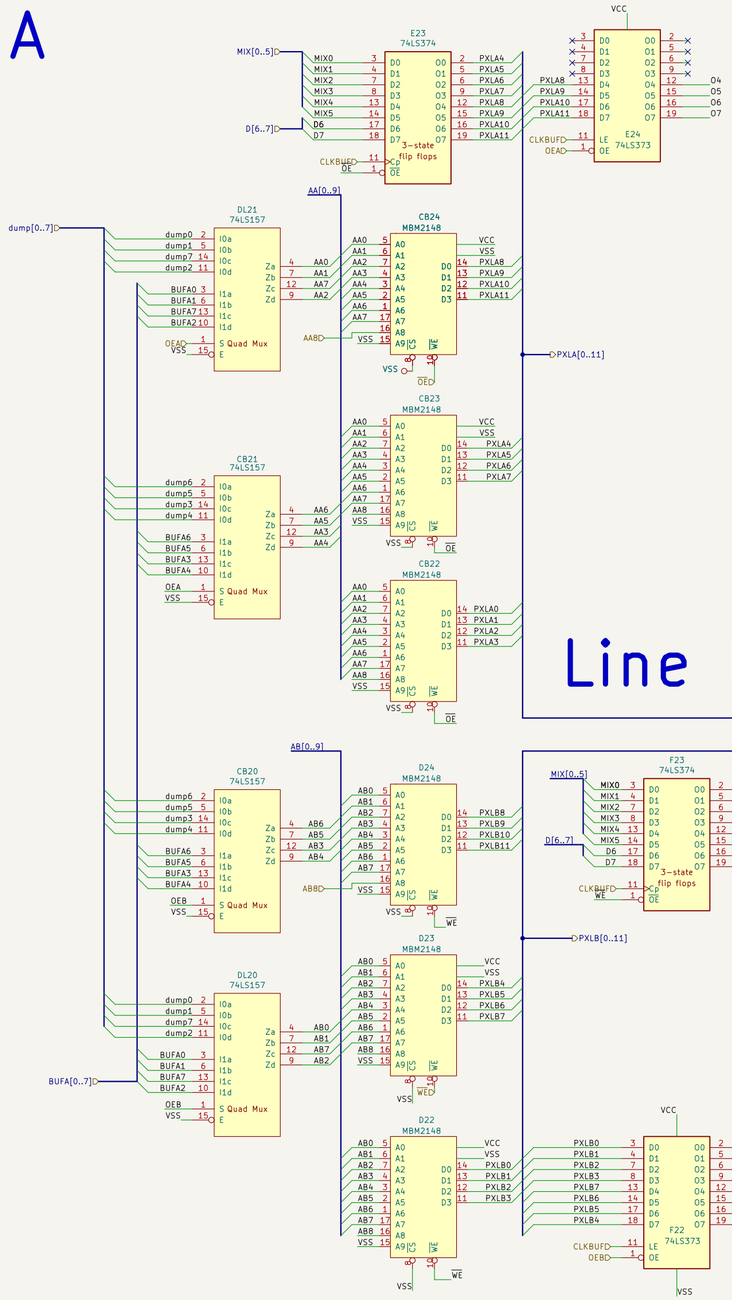


JOTEGO
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Sheet: /boardb/line_counter/
File: linecounter.kicad_sch

Title:

Size: A3	Date: 2022-04-29	**Rev: J. Tejada**
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Line buffer A



Line buffer B

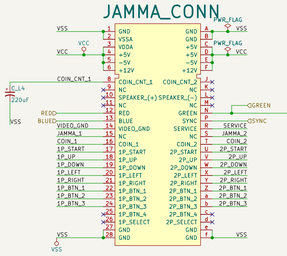
JOTEGO
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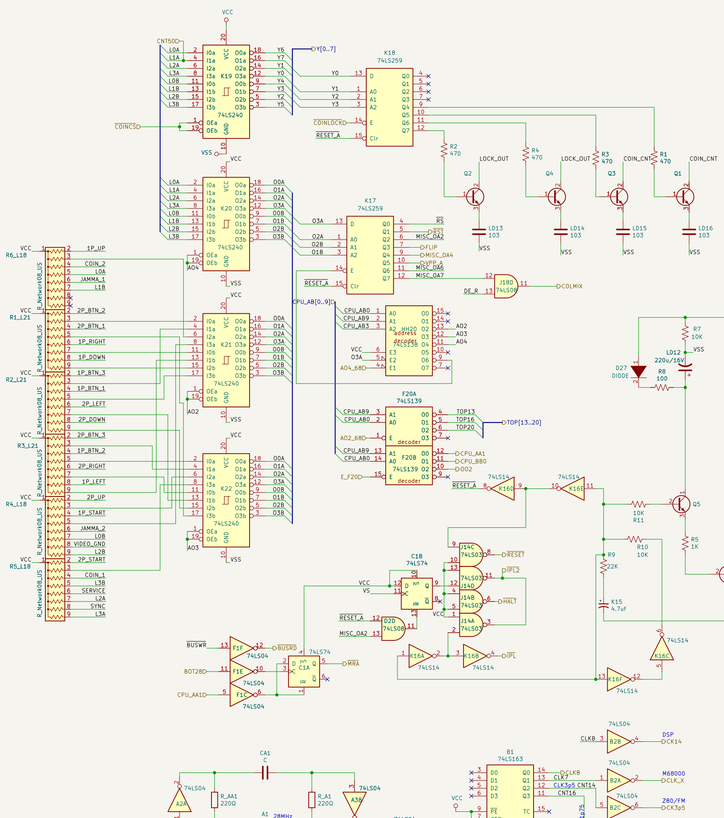
Sheet: /boardb/line_buffer/
File: linebuffer.kicad_sch

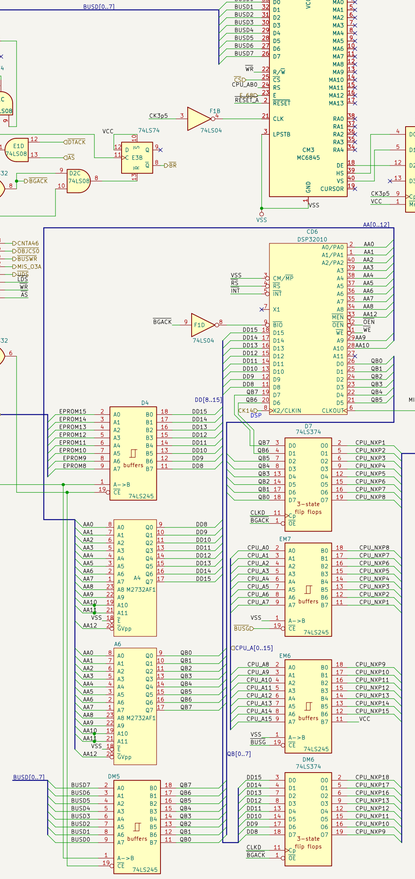
Title: Kyukyoku Tiger –Bootleg–

Size: A2	Date: 2022-04-29	**Rev: J. Tejada**
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JAMMA1
JAMMA_CONN







JOTEGO
JOTEGO

Sheet: /misc/
File: misc.kicad_sch

Title:

Size: A1	Date: 2022-04-29	**Rev: J. Tejada**
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DB[0..15]
VCC
CPU_AB[0..10]
PXL[0..10]
COLMIX_AB
UDB
COL[0..15]
R_WE
VSS
COLMIX
BLUE
GREEN
RED
CLK_XD
CLK_X

FC23 74LS245 — buffers — DB0–DB7 → COL0–COL7
FC22 74LS245 — buffers — DB8–DB15 → COL8–COL15
EC24 CXK5816 — PALA0–PALA10, COL0–COL7
EC22 CXK5816 — PALA0–PALA10, COL8–COL15
DL22 74LS157 — CPU_AB0, CPU_AB9, CPU_AB6, CPU_AB3, PXL0, PXL1, PXL3, PXL2, COLMIX → PALA0, PALA1, PALA3, PALA2
DL23 74LS157 — CPU_AB4, CPU_AB5, CPU_AB2, CPU_AB8, PXL4, PXL5, PXL7, PXL6, COLMIX → PALA4, PALA5, PALA7, PALA6
DL24 74LS157 — CPU_AB7, CPU_AB1, CPU_AB10, PXL8, PXL9, VSS, PXL10, COLMIX → PALA8, PALA9, OE, PALA10
F21F 74LS04 — COLMIX_AB → COLMIX
HC24A 74LS08 — COLMIX
J24A, J24B, HC24B, HC24C, J24D, J24C 74LS08 — COL1–COL6
74LS08 J23B, H23C, J23D, H23D, J23A, H23B, H23A, J23C — COL0–COL7
K24 74LS374 — 3-state flip flops — CLK_XD → BLUE
K23 74LS374 — 3-state flip flops — CLK_X → GREEN
1030 L24
1030 L23
1030 L22

JOTEGO
JOTEGO
Sheet: /colmix/
File: colmix.kicad_sch

Title: Kyukyoku Tiger Bootleg

Size: A3 | Date: 2022-04-29 | **Rev: J. Tejada**
KiCad E.D.A. 10.0.4 | Id: 14/18

CPU_D[0..7]

CPU_A[0..15]

J2
74LS245
buffers
A->B
CE

J1
YM3812

H12
74LS138
address decoder

J21
J20
74LS240

RN_SW2
RN_SW1
R_Network08

SW2_J22
SW1_J19
SW_DIP_x08

Y[0..7]

VSS

HH2
CXK5816

HH3
27256

HH6
Z80CPU

HH1
74LS244

ES5
74LS245

J3A
74LS139
decoder

BUSD[0..7]

K4A
K4B
74LS74

K5C
K5D
K5B
74LS32

R_K3
470ohms1

C103
CP1_Small

C_KK1
10uF

YM3014

HA17558
KL3

YM_L3

C_K1
10uF

FILTER
LK1
uPC1241H

C_K3
47uF

C_L2
220uF

C_L3
10uF

C_LL3
47uF

R_L5
300ohmS1

R_POT_US
RV_K3

RED

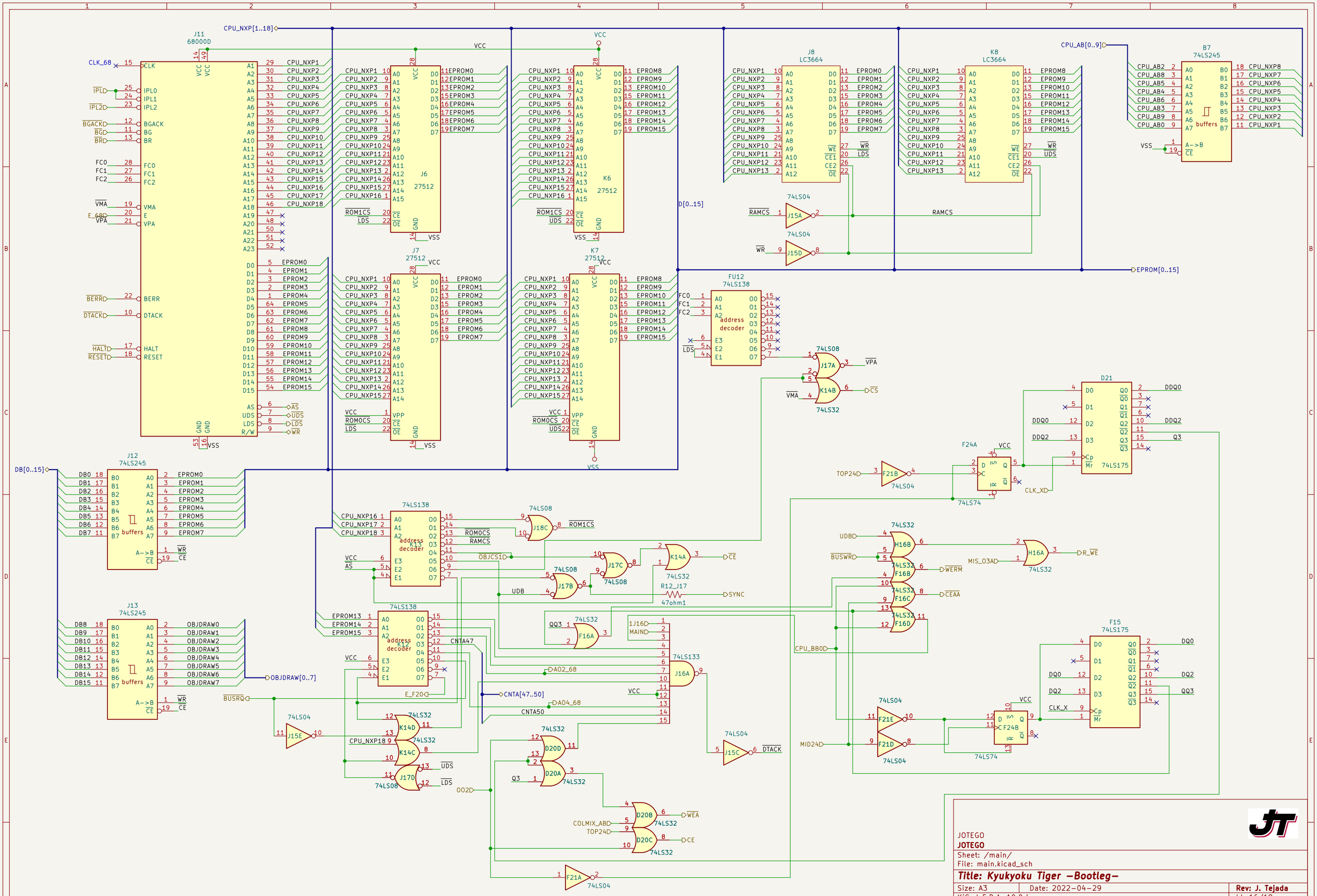
1J16

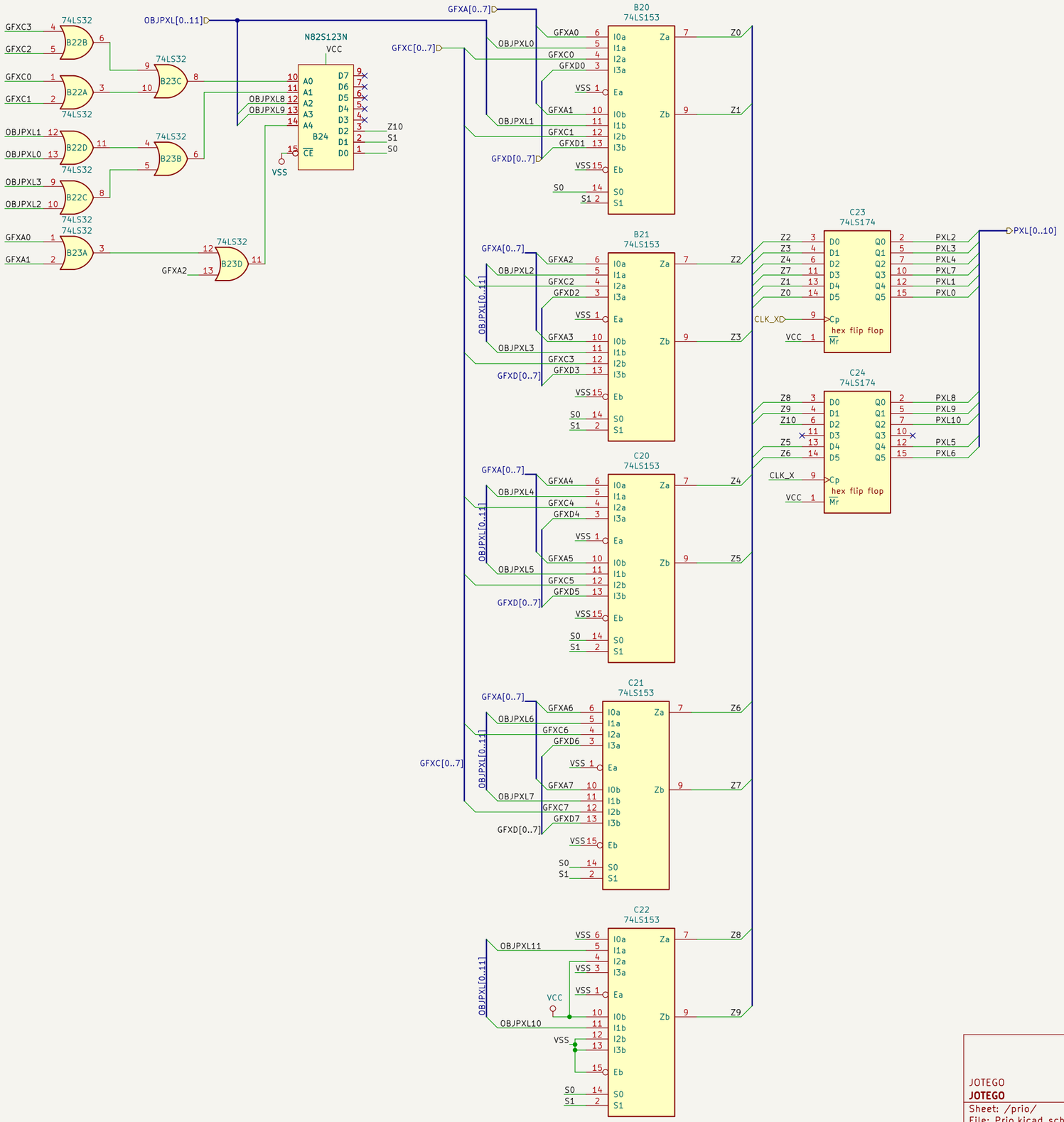
JOTEGO
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Sheet: /sound/
File: sound.kicad_sch

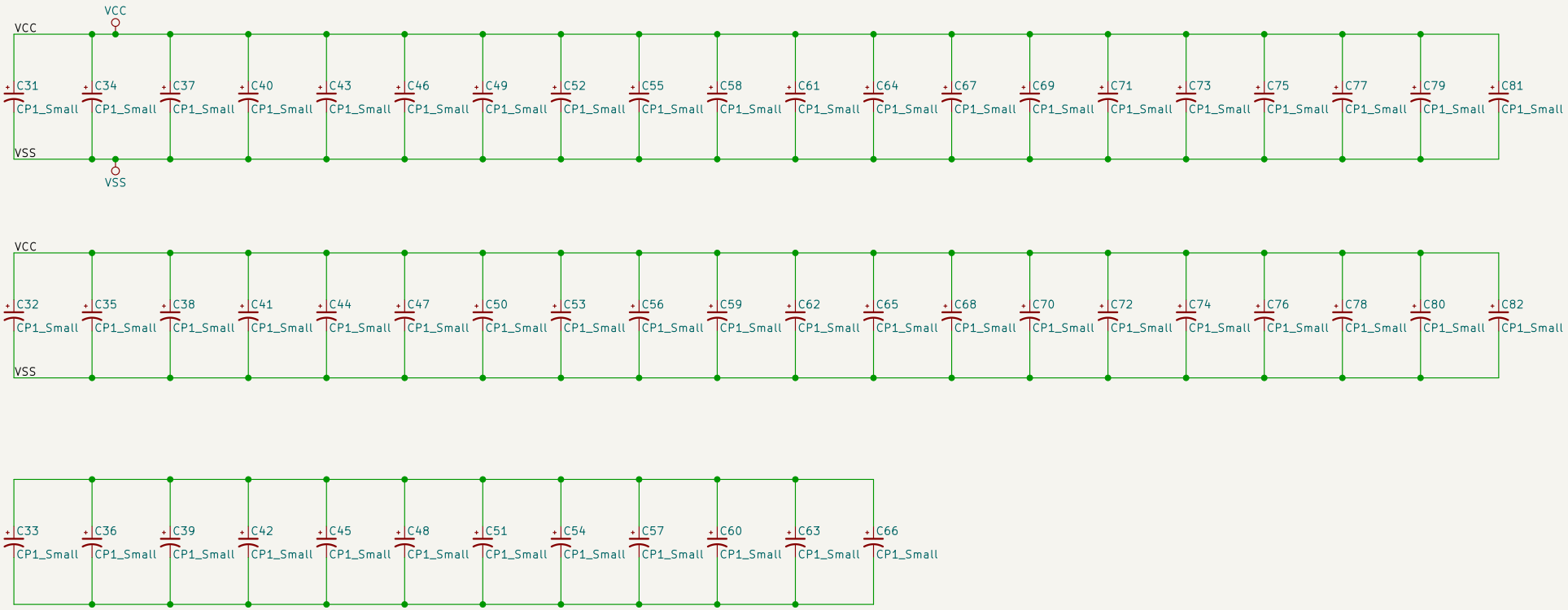
Title: *Kyukyoku Tiger –Bootleg–*

Size: A3	Date: 2022-04-29	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 15/18


CPU_NXP[1..18]
J11
68000D
CLK_68
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E_68
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
VCC
GND
VSS
CPU_NXP1
CPU_NXP2
CPU_NXP3
CPU_NXP4
CPU_NXP5
CPU_NXP6
CPU_NXP7
CPU_NXP8
CPU_NXP9
CPU_NXP10
CPU_NXP11
CPU_NXP12
CPU_NXP13
CPU_NXP14
CPU_NXP15
CPU_NXP16
CPU_NXP17
CPU_NXP18
J6
27512
J7
K6
K7
ROM1CS
ROM0CS
OBJCS1
EPROM0
EPROM1
EPROM2
EPROM3
EPROM4
EPROM5
EPROM6
EPROM7
EPROM8
EPROM9
EPROM10
EPROM11
EPROM12
EPROM13
EPROM14
EPROM15
D[0..15]
J8
LC3664
K8
RAMCS
WR
74LS04
J15A
J15D
CPU_AB[0..9]
B7
74LS245
buffers
A->B
CE
EPROM[0..15]
FU12
74LS138
address
decoder
LDS
74LS08
J17A
K14B
74LS32
VPA
CS
D21
74LS175
DDQ0
DDQ2
Q3
F24A
74LS74
F21B
TOP24
CLK_XD
J12
DB[0..15]
J13
OBJDRAW[0..7]
K13
K12
CNTA[47..50]
A02_68
A04_68
E_F20
BUSRQ
J15E
J15C
J16A
74LS133
1J16
MAIN
CNTA47
CNTA50
QQ3
F16A
F16B
F16C
F16D
H16A
H16B
UDB
BUSWR
MIS_03A
R_WE
WERM
CEAA
SYNC
R12_J17
47ohm1
J17B
J17C
K14A
J18C
K14C
K14D
J17D
D20A
D20B
D20C
F21A
F21D
F21E
F24B
F15
MID24
CPU_BB0
COLMIX_ABD
WEA
OO2
JOTEGO
JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Kyukyoku Tiger –Bootleg–
Size: A3
Date: 2022-04-29
KiCad E.D.A. 10.0.4
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JOTEGO **JOTEGO**		
Sheet: /prio/ File: Prio.kicad_sch		
Title:		
Size: A3	Date: 2022-04-29	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 17/18



JOTEGO
JOTEGO

Sheet: /capacitors/
File: capacitors.kicad_sch

Title:

Size: A4	Date: 2022-04-29	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 18/18